

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
Department of Computer Science and Engineering

**ASSESSMENT TOOL 3– Output Prediction Questions**

|                  |                                                                                                                                                                                                                                                                                                                           |               |                                                          |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------------------------------------------------------|
| Course Code:     | CSA0303                                                                                                                                                                                                                                                                                                                   | Course Title: | Data Structures                                          |
| CO Assessed:     | CO1                                                                                                                                                                                                                                                                                                                       | Assessment:   | Assessment Tool 3 – Debugging and Optimization Questions |
| Weightage:       | 10% of CIA                                                                                                                                                                                                                                                                                                                | Total Marks:  | 100                                                      |
| CO2 Statement:   | Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C', Primitive and non-primitive data structure, Linear and non-linear data structure Arrays & Recursion, Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis. |               |                                                          |
| Student Name:    | ALURI POOJITHA                                                                                                                                                                                                                                                                                                            | Reg. No.:     |                                                          |
| Section / Batch: |                                                                                                                                                                                                                                                                                                                           | Date:         |                                                          |

**Code of Conduct**

I \_\_\_\_\_ POOJITHA \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: \_\_\_\_\_ POOJITHA \_\_\_\_\_

**Instructions**

- This test contains 10 MCQ Based Questions. Total: 100 Marks.
- When a student answer using saveetha compiler, execute the code, Test the test cases , While evaluation the answer should contain following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- Electronic Gadgets are permitted in the class. Take it as test and submit in the saveetha lab compiler class.

**Annexure A – Output Prediction Questions**

1. Write a C program that takes an array as input and returns it reversed.
2. Write a C program for palindrome
3. Write a C program for Searching an Element in a Sorted Array
4. Write a C program for Count the Occurrences of an Element
5. Write a program for binary search
6. Write a C program for Merge Two Arrays
7. Write a C program for Find the Largest and Smallest Element in an Array
8. Write a C program for Find the Second Largest Element
9. Write a C program for Find the First and Last Occurrence of an Element
10. Write a C program for Find Missing Element in an Array

## Rubrics for Calculation

| Criteria                         | Weightage | Level 4 – Exemplary                                                        | Level 3 – Proficient                                  | Level 2 – Developing                           | Level 1 – Beginning                                    |
|----------------------------------|-----------|----------------------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------|--------------------------------------------------------|
| Understanding of Problem Context | 20        | Clearly understands the problem and accurately identifies all requirements | Good understanding with minor gaps                    | Basic understanding; some requirements unclear | Poor understanding or misinterpretation of the problem |
| Application of Concepts          | 30        | Accurately applies appropriate concepts to solve complex problems          | Applies relevant concepts correctly with minor errors | Applies basic concepts with limited accuracy   | Fails to apply appropriate concepts                    |
| Problem-Solving Approach         | 20        | Logical, well-structured, and efficient approach                           | Mostly logical approach with minor gaps               | Basic approach with some inconsistencies       | Illogical or unclear approach                          |
| Accuracy of Solution             | 20        | Completely correct solution with proper justification                      | Mostly correct with minor errors                      | Partially correct solution                     | Incorrect or incomplete solution                       |
| Clarity                          | 10        | Clear, well-organized, and properly explained solution                     | Understandable with minor clarity issues              | Limited clarity and explanation                | Poorly presented and difficult to understand           |
| Faculty In-charge                |           | Course Coordinator                                                         |                                                       | Program Director                               |                                                        |
| Signature: _____                 |           | Signature: _____                                                           |                                                       | Signature: _____                               |                                                        |
| Date: _____                      |           | Date: _____                                                                |                                                       | Date: _____                                    |                                                        |

C++-A13-Output Prediction

Back

### QUESTIONS

Q1

Merging  
10 marks

Q2

Occurrence of an element  
10 marks

Q3

Reverse a list  
10 marks

Q4

Reverse a string  
10 marks

4/4 answered

### Q1 Merging

10 marks

Write a C program for Merge Two Arrays

Your Code

```
1 #include<stdio.h>
2 int main() {
3     int a[5], b[5], c[10], i;
4     for (i=0; i<5; i++) scanf("%d", &a[i]);
5     for (i=0; i<5; i++) scanf("%d", &b[i]);
6     for (i=0; i<10; i++) c[i] = a[i], c[i+5] = b[i];
7     for (i=0; i<10; i++) printf("%d", c[i]);
8     return 0;
9 }
```

Input

12345  
67891

Output

1234567891690-73072030431316710

Run

Save

Submit

## QUESTIONS

Q1

Merging  
10 marks



Q2

Occurrence of an elemnet  
10 marks



Q3

Reverse a list  
10 marks



Q4

Reverse a string  
10 marks



4/4 answered

## Q2 Occurrence of an elemnet

10 marks

Write a C program for Find the First and Last Occurrence of an Element

### Your Code

```
1 #include<stdio.h>
2 int main() {
3     int a[10], i, x, first=-1, last=-1;
4     for (i=0; i<10; i++) scanf("%d", &a[i]);
5     scanf("%d", &x);
6     for (i=0; i<10; i++)
7         if (a[i]==x) {
8             if (first==-1) first=i+1;
9             last=i+1;
10        }
11    if (first==-1) printf("not found");
12    else printf("first=%d last=%d", first, last);
13    return 0;
14 }
```



Run



Save



Submit

### Input

1232452678

### Output

first=2 last=10

## QUESTIONS

Q1

Merging  
10 marks



Q2

Occurrence of an elemnet  
10 marks



Q3

Reverse a list  
10 marks



Q4

Reverse a string  
10 marks



4/4 answered

## Q3 Reverse a list

10 marks

Write a C program that takes an array as input and returns it reversed.

### Your Code

```
1 #include<stdio.h>
2 int main() {
3     int a[5], i;
4     for (i=0; i<5; i++) scanf("%d", &a[i]);
5     for (i=4; i>=0; i--)
6         printf("%d", a[i]);
7     return 0;
8 }
```



Run



Save



Submit

### Input

1 2 3 4 5

### Output

54321

## QUESTIONS

Q1

Merging  
10 marks



Q2

Occurrence of an elemnet  
10 marks



Q3

Reverse a list  
10 marks



Q4

Reverse a string  
10 marks



4/4 answered

## Q4 Reverse a string

10 marks

Write a C program for palindrom

### Your Code

```
1 #include<stdio.h>
2 #include<string.h>
3
4 int main()
5 {
6     char s[5];
7     int i;
8     gets(s);
9     scanf("%s", s);
10    int n=strlen(s);
11    for (i=n-1; i>=0; i--)
12        printf("%c", s[i]);
13    return 0;
14 }
15
16
17
18
```

### Input

maa

### Output

Security Error: Disallowed code pattern detected.